

SAHAANA B

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EDUCATION

SRI KRISHNA COLLEGE OF ENGINEERING AND TECHNOLOGY

BACHELOR OF ENGINEERING (B.E.),
MECHATRONICS ENGINEERING
CGPA: 8.0

SRI SOWDESWARI VIDHYALAYA MATRIC HIGHER SECONDARY SCHOOL

HIGHER SECONDARY EDUCATION
PERCENTAGE: 88

SKILLS

PROGRAMMING

C++
Python
SQL

CORE CONCEPTS

DSA
OOP
DBMS
AI and ML

TOOLS & PLATFORMS

Raspberry Pi
MQTT

CAD TOOLS

SolidWorks
AutoCAD

CERTIFICATIONS

- Introduction to Machine Learning - **NPTEL** (IIT Kharagpur) - Sep 2025
- Wheeled Mobile Robots - **NPTEL** (IIT Madras) - Mar 2025
- AutoCAD Essentials **Course** - SourceCAD - Jan 2025
- Mastering Collaborative Robots **Workshop** - PSG College of Technology, Coimbatore - Sep 2024
- SQL (Basic) **Assessment** - HackerRank - May 2024

EXECUTIVE SUMMARY

MOTIVATED FINAL-YEAR MECHATRONICS ENGINEERING STUDENT WITH HANDS-ON EXPERIENCE IN IoT, EMBEDDED SYSTEMS, AUTOMATION, AND SOFTWARE DEVELOPMENT. PROFICIENT IN C++ AND SQL, WITH PRACTICAL EXPERIENCE IN EMBEDDED PROGRAMMING, SENSOR INTEGRATION, IoT COMMUNICATION, AND REAL-TIME MONITORING SYSTEMS USING ESP32 AND RASPBERRY PI. POSSESSES A STRONG FOUNDATION IN OBJECT-ORIENTED PROGRAMMING, DATABASE MANAGEMENT SYSTEMS, AND ANALYTICAL PROBLEM-SOLVING. SKILLED AT INTEGRATING HARDWARE AND SOFTWARE SOLUTIONS, COLLABORATING EFFECTIVELY IN TEAM ENVIRONMENTS, AND RAPIDLY ADAPTING TO NEW TECHNOLOGIES TO DELIVER INNOVATIVE AND EFFICIENT SOLUTIONS.

EXPERIENCE

ZF SRM CENTRE OF EXCELLENCE | PROJECT TRAINEE

June 2026 | Chennai, Tamil Nadu

- Designed and deployed a full-stack IoT system integrating ESP32 firmware, MQTT broker, and a React.js dashboard for automated hydroponic fertigation control.

VVN ENGINEERING | DESIGN INTERN

May 2026 | Coimbatore, Tamil Nadu

- Developed and modified 3D CAD models and technical drawings to support product design, while gaining hands-on exposure to manufacturing workflows, design validation, and industrial engineering practices.

PROJECTS

DEVELOPMENT OF AN IOT-ENABLED FERTIGATION PREPARATION SYSTEM WITH A BIDIRECTIONAL INTERACTIVE DASHBOARD

May 2026 - June 2026 | Internship Project - SRM University, Chennai

- Built an automated hydroponic nutrient dosing system using ESP32, controlling 4 relay-driven peristaltic pumps with pH (± 0.04 accuracy), TDS/EC, and ultrasonic sensors.
- Developed a bidirectional MQTT - WebSocket - React.js pipeline for real-time monitoring and remote pump control and near-100% message delivery reliability.

SOLAR ELECTRIC CAR - TEAM ASTRA

Aug 2025 - Mar 2026 | Team Project - SKCET, Coimbatore

- Optimized the steering subsystem geometry and linkage using SolidWorks and CAD simulations for stable vehicle handling.
- Strengthened skills in mechanical design, fabrication, testing, and cross-functional teamwork, contributing to **1st Place at the Bharat Solar Vehicle Challenge (Oct 2025)**.